

GNIOT Institute of Management Studies – PGDM (Batch 2023-25)**(TRIMESTER – V) END TERM EXAMINATION, JANUARY -2025****Predictive Analytics and modeling with Python/R****[Time: 2 Hours]****[Maximum Marks: 60]****Following are the course outcomes of the subject: - Predictive Analytics and modeling with Python/R****(Note- Faculty can add the more CO's as per their subject.)**

CO1	Understand and develop practical real-world applications of skills in AI and its application in businesses.
CO2	Design and develop AI modelling techniques using algorithms : supervised and unsupervised
CO3	Enable to collect, examine, and present data with the implementation of AI/predictive modelling algorithms through Python libraries.

SECTION-A**Attempt all parts of the following: (Short Answer Type Questions, do not exceed 150- 200 words)**

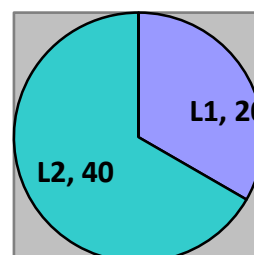
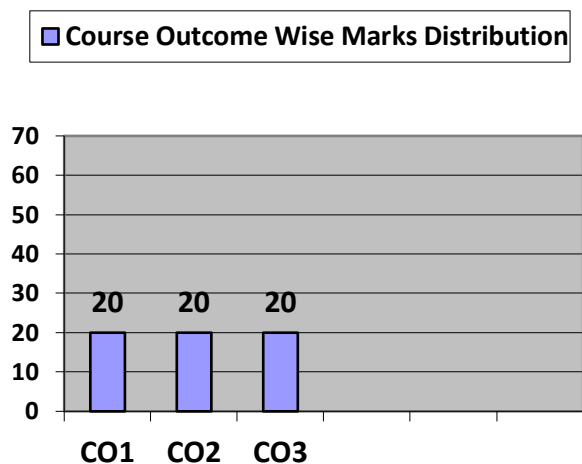
Q.1	Marks	COURSE OUTCOME	BLOOMS LEVEL
(A)	4	CO - 1	BL1
(B)	4	CO - 1	BL1
(C)	4	CO - 1	BL1
(D)	4	CO - 1	BL1
(E)	4	CO - 1	BL1

SECTION-B**Attempt ANY TWO questions out of the following: (Long Answer Type Questions, do not exceed 300- 400 words)**

Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
2 (A)	10	CO – 2	BL2
2 (B)	10	CO – 2	BL2
3 (A)	10	CO – 2	BL2
3 (B)	10	CO – 2	BL2

SECTION-C**Attempt all parts of the following: (Caselet / Numerical, do not exceed 300- 400 words)**

Q.4	Marks	COURSE OUTCOME	BLOOMS LEVEL
4 (A)	10	CO – 3	BL2
4 (B)	10	CO - 3	BL2

**Bloom's Level wise Marks Distribution**

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GNIOT Institute of Management Studies – PGDM (Batch 2023-25)**(TRIMESTER –V) END TERM EXAMINATION, JANUARY - 2025****Predictive Analytics and modeling with Python/R****[Time: 2 Hours]****[Maximum Marks: 60]****INSTRUCTIONS:**

- (i) **There are THREE sections in this question paper.**
(ii) **Attempt questions from each section as per the directions.**
(iii) **Make suitable assumptions wherever necessary.**

SECTION-A**(Short Answer Type Questions, do not exceed 200 words)****Q.1. Attempt all parts of the following:****(5×4=20)**

- (a) Define predictive modelling and its significance in decision-making.
(b) What is the role of feature selection in predictive analysis?
(c) Explain the importance of cross-validation in predictive modelling.
(d) Difference between predictive modelling and predictive analysis.
(e) Difference between dependent and independent variables.

SECTION-B**(Long Answer Type Questions)****Q.2. Attempt all questions of the following****(2X10=20)**

Here is a synthetic dataset with 100 samples of house-related data: Build a linear regression model to predict house prices based on the given dataset.

Size (sq ft)	Number of Bedrooms	Distance to City Center (km)	Price (₹)
3674	1	18.82	930271
1360	5	48.62	424123
1794	5	48.16	1236093
1630	2	13.34	1431565
1595	5	25.37	1129957

2(a) Check for missing values in the dataset and Visualize the relationship between "Size of the house" and "Price of the house" using a scatter plot.

OR

2(b) Compute the correlation matrix for the dataset and Identify which feature has the strongest correlation with "Price of the house."

Q.3. (a) Split the dataset into training (80%) and testing (20%) sets.

OR

(b) Build and train a linear regression model to predict the "Price of the house."

SECTION-C**Q.4. Attempt all parts of the following:****(2×10=20)**

Here is a sample of the generated dataset:

Date	Sales (₹)
2023-01-01	12270
2023-01-02	5860
2023-01-03	10390
2023-01-04	18418
2023-01-05	10191

4(a) Based on the monthly sales data, identify which month has consistently higher sales. Should the store allocate more resources (e.g., inventory, staff) during this high-sales period?

4(b) Should the store implement promotional campaigns to maintain or boost sales above this threshold? Suggest a strategy to increase the number of high-sales days.